



Invention: belief propagating across the reticulum of prior art.

● DEEP RESEARCH LANE • SOURCED & STATE-TAGGED

The Machinery of Invention

Invention is not lightning; it is machinery. Five and a half centuries of patent records — and more than a century of productivity data — show recurring parts, a recognizable process, and measurable output. This paper assembles the elements, the process models, the KPIs, and the historical regularities, then extrapolates only those regularities that have survived measurement.

LANE

Reticulative Logic • Deep Research

METHOD

Web-verified sources • red-teamed draft

EPISTEMICS

Every load-bearing claim state-tagged

DATE

2026-09-03

● TRUTH — verified against cited source

● FALSITY — documented myth / overclaim

◆ IGNORANCE — open; carries a data request

Reticulative convention: claims carry epistemic states, not vibes. Numbers appear only with a citation; where the record is silent, the paper says IGNORANCE and names the data that would settle it.

Invention, innovation, diffusion — three different events

Most confusion about “measuring invention” starts with using one word for three things. The measurement literature separates them precisely.

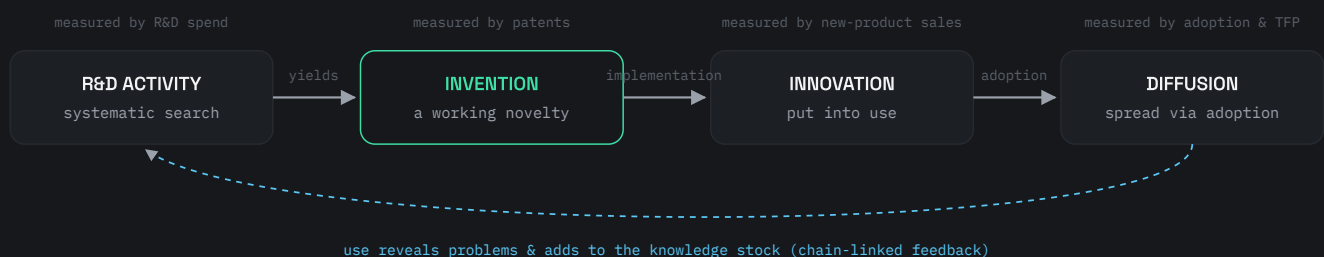
Invention is the creation of a working novelty — a device, process, or method that did not exist before. **Innovation** is that novelty implemented: the OECD’s Oslo Manual, the international standard for innovation statistics, defines an innovation as “a new or improved product or process (or combination thereof) that differs significantly from the unit’s previous products or processes and that has been made available to potential users... or brought into use,” and states the boundary explicitly: *“The requirement for implementation differentiates innovation from other concepts such as invention.”* **TRUTH**

^[1] **Diffusion** is the spread of an implemented innovation through a population of adopters — Everett Rogers’ territory, with its S-curve and adopter categories (innovators 2.5%, early adopters 13.5%, early majority 34%, late majority 34%, laggards 16%).^[2]

The separation is old. Schumpeter built his theory of development on the “carrying out of new combinations” (1911; English ed. 1934)^[3] and in *Business Cycles* (1939) insisted that invention and innovation are distinct events — innovation can occur without new invention, and invention need not induce innovation (paraphrased; the standard citation is vol. 1, p. 84).^[4] The activity that produces inventions is itself standardized for measurement by the OECD’s Frascati Manual (first edition 1963; current 2015): R&D is “creative and systematic work undertaken in order to increase the stock of knowledge... and to devise new applications of available knowledge,” and must be *novel, creative, uncertain, systematic, and transferable and/or reproducible*.^[5]

The three events, and where measurement attaches

Synthesis of Oslo Manual 2018 · Rogers 1962 · Frascati Manual 2015 — drawn for this paper



Each transition loses candidates: most R&D does not yield patentable invention, most inventions are never implemented, most implementations never diffuse widely (\$04 quantifies the attrition). Feedback per Kline & Rosenberg.^[6]

The law's definitions came first, and they encode the same idea: Venice's 1474 statute protected "any new and ingenious device, not previously made"^[7]; England's 1624 Statute of Monopolies reserved 14-year patents to "the true and first inventor" of "new manufactures"^[8]; the US Constitution empowered Congress "to promote the Progress of Science and useful Arts" (1787)^[9], implemented by the Patent Act of 1790.^[10] Novelty, utility, and disclosure have been the legal skeleton of "invention" for five and a half centuries.

02 • THE ELEMENTS

What invention is made of

Ten load-bearing elements, each carried by the historical record rather than by anecdote. Together they are the parts list of the machinery.

Strip away the mythology and the record shows invention behaving like a **process with recurring parts**: a stock of prior art, mechanisms that recombine it, institutions that store, steer, and reward the work, and selection environments that decide what survives. The ten elements below recur from the 15th-century Venetian arsenal to the 21st-century research lab.

E1 • RECOMBINATION

New things are combinations of old things

Brian Arthur's structural analysis: "Technologies somehow must come into being as fresh combinations of what already exists" — components of technologies are themselves technologies. Schumpeter's development was already the "carrying out of new combinations." Invention grows out of the existing stock, which is why it compounds.

Arthur 2009; Schumpeter 1911/1934 ^[11]^[3]

E2 • CUMULATION

Knowledge stacks; nobody starts from zero

Newton to Hooke, Feb 1675/6: "If I have seen further it is by standing on the shoulders of Giants" — itself a 12th-century metaphor (Bernard of Chartres) that Newton was reusing. Every inventor inherits the full prior stack; the stack only grows.

Newton letter, Historical Society of Pennsylvania ^[12]

E3 • INEVITABILITY

Ripe inventions arrive in multiples

Ogburn & Thomas catalogued 148 inventions made independently by two or more people; Merton argued multiples are the norm and singletons the anomaly. Bell's telephone application and Gray's caveat reached the Patent Office the same day — Feb 14, 1876. Darwin and Wallace were read jointly on Jul 1, 1858. When the prerequisites exist, someone assembles them.

PSQ 1922; Merton 1961; patent records [\[13\]](#)[\[14\]](#)[\[15\]](#)[\[16\]](#)

E4 · THE ADJACENT POSSIBLE

Only the next step is reachable

Kauffman's term (1996/2000): at any moment, the space of buildable things is bounded by what exists now; each invention opens new doors. It explains both multiples (many people stand at the same door) and why "ahead of its time" usually means missing complements — calculus needed 17th-century mathematics, not more genius, to be found twice.

Kauffman 1996/2000; Johnson 2010; calculus priority record [\[17\]](#)[\[18\]](#)[\[19\]](#)

E5 · STEERING BY DEMAND & PRICES

Markets point the search

Schmookler: capital-goods patenting tracked investment demand (railroads) — invention follows where buyers are (later re-estimation weakened, but did not erase, the pattern). Hicks: a change in relative factor prices "is itself a spur to invention... directed to economizing the use of a factor which has become relatively expensive."

Schmookler 1966 (+ critique); Hicks 1932 [\[20\]](#)[\[21\]](#)

E6 · INSTITUTIONS

Rules that store, reward, and disclose

Venice 1474, the Statute of Monopolies 1624, the US Patent Act 1790: property in ideas, in exchange for disclosure. The Royal Society (1660) institutionalized open verification. Mokyr's "Republic of Letters" thesis — a competitive market for ideas as the origin of sustained growth — earned half of the 2025 economics Nobel, awarded "for having explained innovation-driven economic growth." Caveat from the record: Moser's world's-fair evidence shows patent laws shaping the *direction* of invention more clearly than its *level*.

Statutes; Royal Society; Mokyr 2016; Nobel 2025; Moser 2005/2013 [\[7\]](#)[\[8\]](#)[\[10\]](#)[\[22\]](#)[\[23\]](#)[\[24\]](#)[\[25\]](#)

E7 · ORGANIZATION

Invention industrialized itself

Edison's Menlo Park (1876) promised "a minor invention every ten days and a big thing every six months or so." GE built the first US industrial lab (1900); Bell Labs consolidated at ~3,600 staff (1925). Individual inventors out-patented corporations until 1933; today ~86% of US-origin utility patents go to businesses. The unit of invention moved from person to system.

E8 · PREPARED SERENDIPITY

Accidents reward the trained observer

Pasteur, Douai, Dec 7, 1854: "chance favors only the prepared mind." Fleming's contaminated plate (Sep 3, 1928), Goodyear's hot stove (1839; patent 3,633, 1844), Spencer's melted chocolate bar at a magnetron (1945), Röntgen's glowing screen (Nov 8, 1895): each accident happened in front of someone equipped to recognize it — and inside an institution able to develop it.

Wikisource 1854; ACS; USPTO; IEEE Spectrum; APS [31][32][33][34][35][36]

E9 · VARIATION & SELECTION

Technology evolves; markets select

Basalla: continuity, diversity, novelty, selection — an evolutionary account of artifacts. Nelson & Winter: firms carry routines like genes; innovation is search; markets are the selection environment. Invention supplies variation; selection decides what becomes innovation.

Basalla 1988; Nelson & Winter 1982 [37][38]

E10 · OPEN EXCHANGE

Sharing accelerates the whole district

Allen's "collective invention": Cleveland-district (England) ironmasters freely revealed furnace designs, and the whole industry ratcheted forward for ~20 years. The Royal Society's journals, Mokyr's Republic of Letters, and modern open source are the same mechanism at different scales: disclosure compounds the stock everyone recombines from (E1, E2).

Allen 1983; Royal Society; Mokyr 2016 [39][22][23]

[TRUTH] Multiples are the documented norm – 148 catalogued cases by 1922; same-day telephone filings; joint Darwin-Wallace reading. [13][15][16]

[TRUTH] The incandescent lamp had "nearly a dozen" active inventors in the 1870s before Edison's carbon-filament system won (Friedel & Israel; US DOE). [40]

[TRUTH] Team share of knowledge production rose across five measured decades; mean S&E team size grew from under 2 to ~3.5 authors by 2000. [41]

[FALSITY] "Great inventions are the work of solitary geniuses." Lemley's review of the legal-historical record: nearly all significant inventions were made near-simultaneously by multiple teams. [42]

=> The lone-genius account: **FALSITY** – individuals matter, but as the fastest assemblers of a ripe adjacent possible, not as sole causes.

03 · THE PROCESS

A century of process models, converging

The models of how invention happens were themselves invented, tested, and revised. Read in sequence, they converge on one architecture: staged, feedback-ridden, and selective.

Every serious process model since 1926 has kept **stages** and added **feedback**. Wallas named the cognitive stages; Usher embedded them in cumulative history; the postwar "linear model" (basic research → applied → product) turned out to be a retrospective construction that its supposed author never drew; Kline and Rosenberg replaced it with chains and loops; Cooper turned the loops into managed gates; Altshuller showed the patterns could be mined from the patent record itself.

1926

Wallas — the four stages

Preparation → Incubation → Illumination → Verification (with an "Intimation" sub-stage, per later re-reading). The cognitive core every later model keeps. [43]

1929 / 1954

Usher — cumulative synthesis

Perception of the problem → setting the stage → act of insight → critical revision. Insight is real but bracketed by preparation and by the long grind afterwards. [44]

1943 / 1962

Ryan & Gross → Rogers — diffusion

The Iowa hybrid-corn study (259 farmers, near-total adoption in ~13 years) founded diffusion research; Rogers systematized the S-curve and adopter categories.^{[47][2]}

JUL 1945

Bush — Science, the Endless Frontier

Basic research as “the pacemaker of technological progress”; the report that built the NSF — and got retrofitted into a “linear model” of innovation.^[45]

1946

Altshuller — TRIZ

Invention patterns mined from ~40,000 patent abstracts: 40 inventive principles, contradiction matrix. The first systematic demonstration that the inventive step itself has regularities. (Larger patent counts claimed by the TRIZ community are undocumented.)^[46]

1986

Kline & Rosenberg — the chain-linked model

“Models that depict innovation as a smooth, well-behaved linear process badly misspecify the nature and direction of the causal factors at work.” Research is called on all along the chain; use feeds back into design.^[6]

1986 / 1990

Cooper — Stage-Gate

The funnel operationalized: staged development with Go/Kill gates — selection built into the process rather than left to the market’s end.^[48]

2006

Godin — the linear model debunked

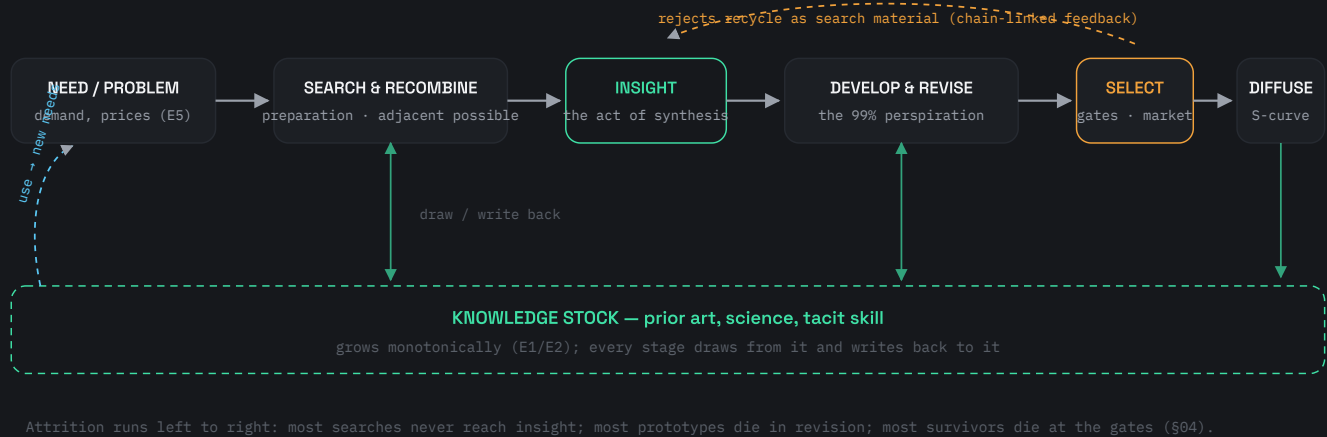
Historiography: the “linear model” was assembled retrospectively in three steps by different communities. The tidy pipeline was never anyone’s empirical finding.^[49]

The consensus architecture

Overlaying the surviving models yields one canonical loop — the process this paper measures in §04 and extrapolates in §06:

The canonical invention loop

Synthesis drawn for this paper from Wallas 1926 · Usher 1954 · Kline & Rosenberg 1986 · Cooper 1990 · Rogers 1962 – not any single author's diagram



Edison's ratio – "Genius is 1 per cent inspiration and 99 per cent perspiration" – is documented from 1901 and describes the DEVELOP & REVISE box, where the process spends most of its time.^[50]

The demand side has stages too

Diffusion is not an afterthought: it is where invention becomes economically visible, and it has measured structure. Rogers' adopter distribution:^[2]

Rogers' adopter categories — share of eventual adopters

Diffusion of Innovations (1962); percentages are Rogers' idealized normal-curve segments



Single-hue ordinal ramp (adoption order); labels carry identity. Source: Rogers 1962.^[2]

04 · THE MEASUREMENTS

KPIs and metrics: what the dashboard actually shows

Invention is measured as a pipeline: inputs (money, people) → activity (R&D) → outputs (patents, papers, products) → outcomes (productivity). Every number below is the latest published value; bases are named because mixing them is the #1 measurement error.

GLOBAL R&D SPEND · 2024

\$3.48T

current PPP dollars; China 30%, US 29% of world

NSF NCSES, State of US S&E 2026 [\[51\]](#)

PATENT APPLICATIONS · 2024

3.7M

+4.9% — fastest growth since 2018; Asia files 70.1%

WIPO WIPI 2025 [\[52\]](#)

PATENTS IN FORCE · 2024

19.7M

grants ran ~2.1M in 2024; US cumulative issuance passed 12M in June 2024

WIPO WIPI 2025; USPTO [\[52\]](#) [\[53\]](#)

S&E ARTICLES · 2022

3.3M

up from 2.0M (2010); China's output >2× the US by 2023; US leads in highly-cited share

NSF S&E Indicators [\[54\]](#)

VENTURE FUNDING · 2025

\$425B

Crunchbase basis (+30% YoY); PitchBook basis \$512.6B — AI took ~half of either

Crunchbase EOY 2025; PitchBook [\[55\]](#) [\[56\]](#)

RESEARCHERS · 2018

8.85M

FTE worldwide; researchers per million rose 1,141 (2015) → 1,486 (2023)

UNESCO Science Report 2021; UIS [\[57\]](#) [\[58\]](#)

Basis discipline — the same world, three totals

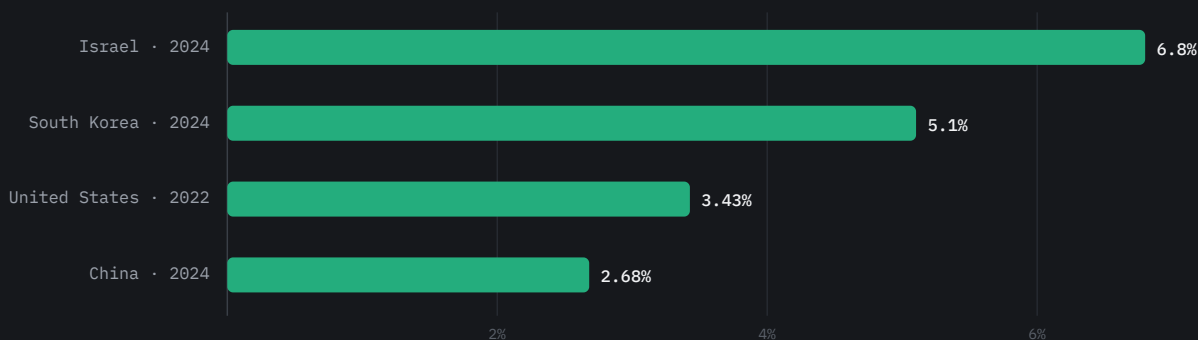
- Global R&D 2024 is **\$3.48T in current PPP dollars** (NSF)^[51] and **\$2.87T in constant-2015 PPP dollars** (WIPO's tracker)^[59] — both correct, different bases. Never mix them in one series.
- China vs US: in PPP terms China's R&D (**\$1.028T**) edged past the US (**\$1.009T**) in 2024^[51]; at market exchange rates China remains roughly half the US level (OECD, March 2026 release).^[60] Which "truth" you get depends on the conversion you choose.
- Totals that look interchangeable rarely are: 2025 global venture funding is **\$425B** under Crunchbase's methodology and **\$512.6B** under PitchBook's — an ~\$88B gap from inclusion rules alone (both cited at the tile above). Name the methodology with the number.

Input metrics

The workhorse input KPIs are **GERD** (gross domestic expenditure on R&D), **R&D intensity** (GERD/GDP), **researchers per million**, and **risk capital**. The US spent \$937B in 2023 (est. \$993B in 2024) — split \$138B basic research, \$174B applied, \$625B experimental development^[61] — with business funding 75% and the federal government 18%, a near-inversion of 1964 (business 31%, federal 67%).^[62] China reported ¥3.61T for 2024 (+8.3%, intensity 2.68%) and ¥3.93T for 2025 (intensity 2.8%).^[63] Government R&D budgets across the OECD fell 4.1% in real terms in 2024, with defence the main rising category.^[60]

R&D intensity — who commits the largest share of their economy

GERD as % of GDP · years differ by source and are labeled per bar



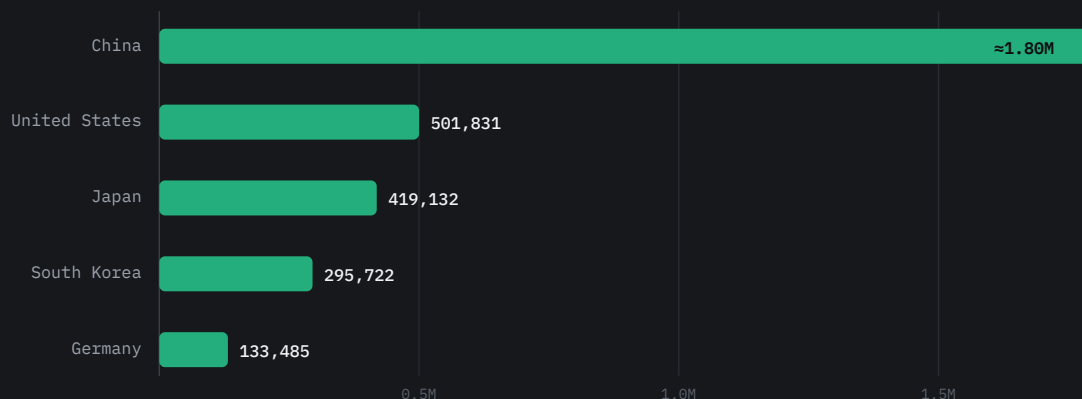
Sources: OECD MSTI Mar 2026 (Israel, Korea)^[60]; NSF (US, 2022 — latest finalized ratio)^[64]; China NBS (2024).^[63] Mixed vintages disclosed; gaps are larger than vintage drift.

Output metrics

Patents are the oldest continuous output series — the US register alone runs from Samuel Hopkins’ potash patent (July 31, 1790) to ~12.7 million numbered patents by August 2026.^[53] The 2024 world totals: 3.7M applications (+4.9%), ~2.1M grants, 19.7M in force; 11.7M trademark applications; 1.22M industrial-design applications.^[52] The PCT route — one international application — ran 275,900 filings in 2025, led by China (26.7%) and the US (19.1%).^[65]

Patent applications by origin, 2024

Resident + abroad filings by origin · WIPO World Intellectual Property Indicators 2025



Source: WIPO WIPI 2025 (Nov 18, 2025).^[52] Counts confound inventive output with propensity-to-patent — see the pathology panel below.

Outcome metrics — where invention meets the economy

The outcome layer is productivity. Solow’s 1957 decomposition attributed **87.5%** of the 1909–1949 doubling of US output per hour to technical change^[66] — a residual Solow defined broadly (“any kind of shift in the production function”), but still the founding demonstration that capital accumulation alone cannot explain long-run growth. The modern readings are sobering: US TFP growth ran ~1.98%/yr in 1948–1972 but only ~0.87%/yr across 1973–2019, and post-2004 sits below even the 1973–96 slowdown^[67]^[68]; Gordon puts post-1970 TFP growth at roughly one-third of the 1920–1970 rate.^[69]

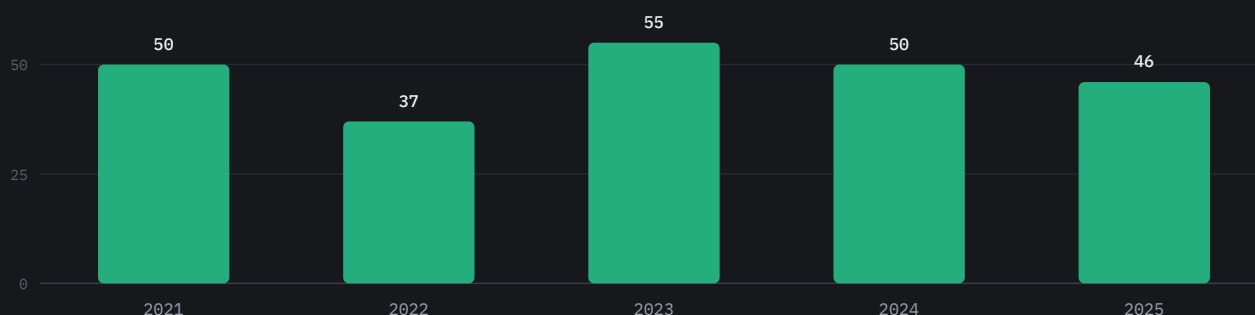
A natural experiment in metric behavior: pharma

Drug R&D is the best-instrumented invention pipeline on earth, and it shows both metric pathologies and real signal. **Eroom’s Law** (Moore’s, backwards): approved drugs per inflation-adjusted billion dollars of R&D *halved roughly every 9 years from 1950 to 2010* — an ~80-fold efficiency decline^[70] — before breaking upward around 2010.^[71] Cost per

approved drug: \$2.558B pre-approval (Tufts, 2013 dollars)^[72]; Deloitte's cohort average ran \$2.23B with 5.9% IRR in 2024, improving to \$2.67B with 7.0% IRR in 2025.^[73]

FDA novel drug approvals (CDER), 2021–2025

Novel approvals per calendar year – the pipeline's output counter



Source: FDA CDER novel approvals; 2025 summary via RAPS.^[74] Output holds ~37–55/yr while cost-per-asset and returns move – a reminder that count metrics and efficiency metrics answer different questions.

Firm-level KPIs — the practitioner's panel

Corporate R&D concentrates hard: the world's top 2,000 corporate investors spent **€1,446B in 2024** — over 90% of all business-funded R&D — with US firms carrying 47.1% of it; Amazon entered the ranking at #1 (~€65B, a Scoreboard estimate), and FY2024 filings show Alphabet at \$49.3B and Meta at \$43.9B.^[75]^[76] The standard firm KPI panel:

- **R&D intensity** (R&D / net sales): pharma & biotech lead at ~16%, software & computer services ~12% (EU Scoreboard sector data; edition vintages vary). **IGNORANCE** — current-edition sector split not verified; treat the levels as indicative. ^[75]
- **New Product Vitality Index** (3M, 1988): share of revenue from products launched in the past five years; 3M's benchmark ~30%. ^[77]
- **RQ (Research Quotient)**: Knott's estimated output elasticity of R&D — the closest thing to a firm-level "return on invention."^[78]
- **Funnel metrics**: stage-gate pass rates, time-to-market (no standardized definition — start and end points vary by firm), patents per R&D dollar. ^[48]
- **Composite indices**: the Global Innovation Index 2025 (139 economies) ranks Switzerland #1 for the 15th consecutive year; China entered the top 10 for the first time. GII 2026 is due Sep 29, 2026 — after this paper. ^[79]

[TRUTH] Inputs and formal outputs are measured to international standard (Frascati 2015; Oslo 2018; WIPO series back to the 19th century). [\[5\]](#)[\[1\]](#)[\[52\]](#)

[TRUTH] Output metrics are gameable and propensity-confounded: firms rank secrecy and lead time above patents as protection (1,478-lab survey); and wherever counts become targets, gaming follows – "when a measure becomes a target, it ceases to be a good measure" (Strathern's formulation of Goodhart's law). [\[80\]](#)[\[81\]](#)

[FALSITY] "Patent counts measure invention rates." They measure filings: only ~5% of US patents are ever litigated or licensed for a royalty (~1.5% litigated), and "perhaps 95%" are never valuable enough for either (Lemley, ca. 2001). [\[82\]](#)

[IGNORANCE] The *current* patent→value conversion rate.

DATA REQUEST: a post-2001 replication of Lemley's licensing/litigation rates on modern USPTO cohorts.

=> Invention is measurable – **as a battery of imperfect gauges** (inputs + outputs + outcomes), never as any single number.

05 · THE LESSONS

Thirteen things history actually established

Each lesson is stated as a claim, with its evidentiary basis and epistemic state.

These thirteen are the premises the trajectory section is allowed to extrapolate from.

01

Invention is recombination, so it compounds **TRUTH**

New technologies are combinations of existing ones; each addition enlarges the space of possible next combinations. This is why the process accelerates across centuries rather than exhausting itself.

Arthur 2009; Schumpeter 1911/1934; Kauffman's adjacent possible [\[11\]](#)[\[3\]](#)[\[17\]](#)

02

Ripe inventions arrive on schedule, in multiples **TRUTH**

148 documented independent co-inventions by 1922; same-day telephone filings (Feb 14, 1876); Darwin–Wallace read jointly (1858). When prerequisites and demand are present, the invention is close to inevitable — the interesting variable is *who assembles it first*, not whether.

Ogburn & Thomas 1922; Merton 1961; Lemley 2012 [\[13\]](#)[\[14\]](#)[\[42\]](#)

03

The lone genius is bad history **FALSITY of the myth**

The canonical solo inventors dissolve on inspection (“nearly a dozen” incandescent-lamp inventors in the 1870s), and knowledge work has shifted decisively to teams: mean S&E team size grew from under 2 to ~3.5 authors in 45 years, and teams produce the most-cited work.

Lemley 2012; US DOE lamp history; Wuchty, Jones & Uzzi 2007 [42][40][41]

04

Institutions are the operating system **TRUTH**

Patents (1474 → 1624 → 1790) bought disclosure with temporary exclusivity; open-verification societies (1660) made knowledge cumulative and trustworthy; the 2025 economics Nobel was awarded precisely “for having explained innovation-driven economic growth” through institutions and creative destruction. Nuance: patent laws demonstrably steer invention’s *direction*; their effect on its *level* is contested.

Venice 1474; Statute of Monopolies; Royal Society; Nobel 2025; Moser 2005/2013 [7][8][22][24][25]

05

Invention industrialized: person → lab → team → system **TRUTH**

Menlo Park scheduled invention (“a minor invention every ten days”); GE (1900) and Bell Labs (1925) made it a corporate function; individual inventors out-patented firms until 1933, while ~86% of US-origin utility patents now go to businesses. The production function of invention is an organizational variable, not a constant.

Rutgers Edison Papers; IEEE Milestones; Smithsonian/NSF shares; Lamoreaux & Sokoloff [26][27][28][29][30]

06

Demand and relative prices steer the search **TRUTH – direction; magnitude debated**

Railroad patenting tracked railroad investment (Schmookler); dear factors invite economizing inventions (Hicks 1932). Re-estimation weakened Schmookler’s correlations, so the lesson is directional steering, not demand-determinism.

Schmookler 1966 + Research Policy critique; Hicks 1932 [20][21]

07

Impact arrives decades after invention **TRUTH**

Electric motors were <5% of US factory drive power in 1899 — 17 years after Edison’s first central station — and passed 50% only in the early 1920s, when the productivity surge finally appeared. Solow’s 1987 quip (“the computer age everywhere but in the productivity statistics”) marked the same lag for IT; the J-curve formalizes it via intangible complements.

David 1990; Solow 1987; Brynjolfsson, Rock & Syverson 2021 [83][84][85]

08

Science and invention have fused **TRUTH**

Patents citing science rose from 6.7% (1976) to 25.6% (2018); 61% of US patents trace back to a research article and 80% of cited articles link forward to a patent.

Meanwhile large firms retreated from doing science themselves (publishing firms 17%→6%, 1980–2006) — buying the goose’s eggs while releasing the goose.

Marx & Fuegi 2020; Ahmadpoor & Jones 2017; Arora, Belenzon & Pataconi 2018 [86][87][88]

09

Each idea costs more than the last TRUTH – with a live dissent

Aggregate US research productivity fell $\times 41$ since the 1930s ($\approx -5.1\%/yr$); a Moore’s-law doubling now takes $>18\times$ the researchers of the early 1970s; age at first invention rose ~ 0.6 years/decade as the knowledge burden grew; drug R&D efficiency halved every ~ 9 years for six decades before breaking upward around 2010. Dissent exists (Census WP 25-29: patents per R&D input rising) and is flagged in §06.

Bloom et al. 2020; Jones 2009; Scannell 2012; Ringel 2020; Census 2025 [89][90][70][71][91]

10

Growth held because effort scaled TRUTH

The response to falling research productivity was more researchers and more money: global R&D reached \$3.48T (2024), and in PPP terms China’s spend passed the US in 2024 — the first change at the top since the series began. Scaling inputs is the historical answer to declining idea-yield.

Bloom et al. 2020 (effort doubling); NSF 2026 [89][51]

11

Cost curves are the most predictable regularity we have TRUTH

Wright’s 1936 aircraft learning curve generalizes: across 50+ technologies, cost declines follow statistically forecastable exponentials. Solar modules fell \$106/W (1976) → $\sim \$0.38/W$ ($\sim 99.6\%$), at a 20.2% learning rate per doubling of cumulative capacity. Where a deployment-driven curve exists, forecasting invention’s *effects* becomes tractable even when individual inventions stay unpredictable.

Wright 1936; Farmer & Lafond 2016; Way et al. 2022; OWID [92][93][94][95]

12

Serendipity is a harvest, not a lottery TRUTH

Penicillin, vulcanization, X-rays, the microwave: each “accident” landed on a trained observer inside an institution able to develop it. Pasteur said it in 1854; the 20th-century lab systematized it. Slack time and exploratory capacity are inputs, not waste.

Pasteur 1854; Fleming 1928; Röntgen 1895; Spencer 1945; 3M Post-it [31][32][35][34][36]

13

Openness accelerates the whole district TRUTH

Cleveland’s ironmasters ratcheted an entire industry by publishing furnace designs; the Republic of Letters did it for Enlightenment Europe; patents themselves are a

disclosure bargain. Sharing enlarges the common stock that everyone recombines from — the compounding term in Lesson 1.

Allen 1983; Mokyr 2016; Royal Society [\[39\]](#)[\[23\]](#)[\[22\]](#)

06 • THE TRAJECTORY

Estimating forward without inventing the future

This section extrapolates only the regularities established in §05, reports published estimate ranges where experts disagree, and names the data that would falsify each projection. It is an extrapolation, not a prophecy.

6.1 • The scissors: more effort required, fewer hands available

Two measured curves are closing on each other. The first is research productivity: holding growth constant has required doubling research effort every ~13 years.[\[89\]](#) The second is demography: the researcher pipeline ultimately draws on population, and the UN projects world population to peak around 10.3 billion in the mid-2080s — it has already peaked in 63 countries.[\[96\]](#) Chad Jones’ growth theory makes the collision explicit: with sub-replacement fertility, idea-based growth models converge to an “Empty Planet” outcome of stagnating living standards.[\[97\]](#)

RESEARCH PRODUCTIVITY SINCE 1930S

÷41

≈ -5.1%/yr aggregate US decline

Bloom, Jones, Van Reenen & Webb, AER 2020 [\[89\]](#)

EFFORT NEEDED TO HOLD GROWTH

2× / 13y

research effort must double every 13 years

Bloom et al. 2020 [\[89\]](#)

MOORE'S-LAW UPKEEP

>18×

researchers per chip-density doubling vs early 1970s

Bloom et al. 2020 [\[89\]](#)

10.3B

projected mid-2080s; 63 countries already past peak

UN WPP 2024 ^[96]

Worked comparison (derived here, shown in full): the 13-year doubling requirement implies effort growth of $2^{1/13} - 1 \approx 5.5\%/yr$. UNESCO's series shows researchers per million rising 1,141 (2015) $\rightarrow$ 1,486 (2023)^[58]: a ratio of 1.302 over 8 years, i.e. $1.302^{1/8} - 1 \approx 3.4\%/yr$. The per-capita series is thus growing ~ 2 points/yr slower than the requirement.

IGNORANCE The precise absolute-headcount gap.

DATA REQUEST: UNESCO UIS absolute FTE researcher series 2015–2023 (latest verified point: 8.854M FTE in 2018^[57]) to compute total-effort CAGR against the 5.5%/yr requirement.

A live dissent must be carried, not buried: a 2025 Census Bureau working paper finds patents per R&D input *rising* — "Ideas Are Not Getting Harder to Find."^[91] The disagreement is about how to measure idea output; \$06's projections therefore lean on the scissors only for direction, never for a point value.

6.2 • The counterforce: invention of new methods of invention

The historical precedent is exact. Whitehead, 1925: "The greatest invention of the nineteenth century was the invention of the method of invention."^[98] Griliches showed hybrid corn was "the invention of a method of inventing" (1957)^[99]; Cockburn, Henderson & Stern argued in 2018 that deep learning is the next general-purpose method of invention.^[100] Since then the evidence has moved from argument to record:

NOV 2020

AlphaFold 2 solves the CASP14 benchmark

Recognized by CASP organizers as a solution to the 50-year protein-structure problem; Nature paper Jul 2021.^{[101][102]}

JUL 2022

The protein universe, predicted

AlphaFold DB expands to 214,684,311 structures — "nearly all catalogued proteins known to science."^[103]

NOV 2023

Materials at scale — with a caution

GNoME predicts 2.2M crystals (~380k stable; 736 realized concurrently).^[104] The 2024 *Chemistry of Materials* critique found "scant evidence" of compounds meeting "the trifecta of novelty, credibility, and utility" — scale is not yet usefulness.^[105] GraphCast beats the leading physics weather model on ~90% of 1,380 targets the same month.^[106]

OCT 2024

The Nobel threshold

Chemistry Nobel: half to Baker "for computational protein design," half jointly to Hassabis and Jumper "for protein structure prediction." AI-assisted invention enters science's highest canon.^[107]

FEB–MAY 2025

Hypothesis machines

Google's multi-agent "co-scientist" reproduces an unpublished, experimentally confirmed superbug mechanism among its top-ranked hypotheses.^[108]

AlphaEvolve improves on Strassen's 1969 matrix-multiplication record (48 multiplications for 4×4 complex), recovers ~0.7% of Google's fleet compute, and improves best-known solutions on ~20% of ~50 open problems.^[109]

FutureHouse's Robin proposes a drug-repurposing candidate with humans running the wet lab.^[110]

JUN 2025

First generative-AI-discovered drug clears Phase IIa

Rentosertib (generative-AI target + molecule) publishes randomized Phase IIa results in *Nature Medicine*: +98.4 mL FVC (60 mg) vs -20.3 mL placebo in IPF; Phase III began Jul 2026.^{[111][112]} As of mid-2026, **no end-to-end AI-discovered drug is FDA-approved.**^[113]

JUL 2025

IMO gold, certified

Gemini Deep Think scores 35/42 — gold-medal standard, officially graded by IMO coordinators (OpenAI claimed the same score, graded privately, uncertified). One year earlier AI stood at silver; two years earlier, at geometry-only.^{[114][115][116]}

MAR 2026

The referee joins the loop

Sakana's AI-Scientist line — whose workshop paper passed peer review in 2025 under organizer supervision — publishes in *Nature*, including an automated reviewer that matches human judgments. Generation *and* verification are both automating.^[117]

JUL 2026

Reported, not yet established

IGNORANCE Press reports say multiple AI systems scored perfect 42/42 at IMO

2026 (Shanghai). Secondary reporting only; not independently confirmed for this paper. DATA REQUEST: official IMO 2026 grading records.^[118]

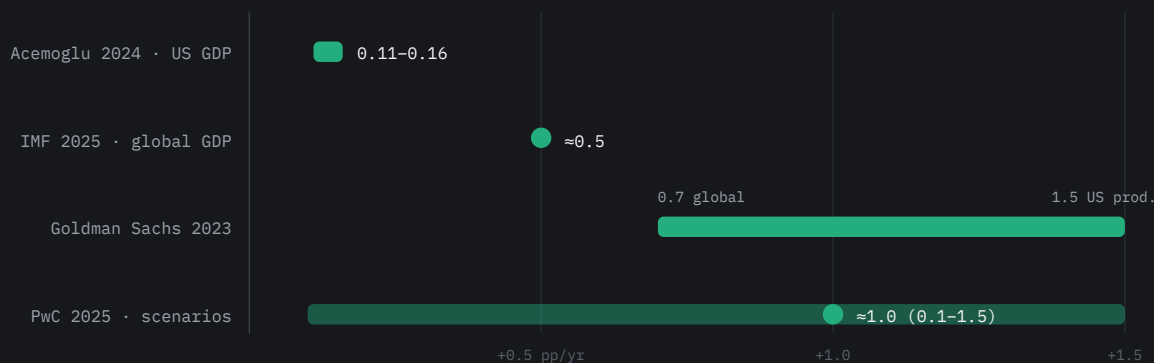
Note what the timeline does *not* show: an approved AI-discovered drug, an AI-designed material in mass production, or an autonomous system running the full loop of \$O3 — including the wet lab and the market gate — without humans. The pattern so far matches \$O5's Lesson 7: the SEARCH and INSIGHT boxes are automating fastest; DEVELOP-and-REVISE and SELECT still run at human and regulatory speed. That is exactly where electrification stood in 1899 — capability demonstrated, complements pending. **TRUTH as description; the electrification analogy is an interpretive frame, not evidence**

6.3 • What the credible estimates disagree about

Institutions that have modeled AI's aggregate effect disagree by roughly an order of magnitude. Rather than pick a winner, the paper plots the published range, normalized to an approximate annualized boost (derivations in the table):

Published estimates of AI's boost to annual growth

Approximate percentage points per year over each study's ~10-year horizon • scopes differ and are labeled



Sources: Acemoglu^[119] • IMF via WEF (secondary)^[120] • Goldman Sachs^[121] • PwC^[122]. McKinsey's \$2.6–4.4T/yr is a level effect once adopted, not a growth rate, so it is excluded from this axis.^[123] Scopes differ (US vs global; GDP vs labor productivity) — this is a disagreement range, not a distribution.

6.4 • The input curves the projections ride on

FRONTIER TRAINING COMPUTE

4–5×/yr

growth 2010–2024 $\approx 4.1\times/\text{yr}$; $\sim 4.4\times/\text{yr}$ since 2022

Epoch AI [\[124\]](#)

AI TASK-LENGTH DOUBLING

~ 7 mo

human-time length of tasks agents complete at 50% success, ~ 6 -yr trend

METR 2025 [\[125\]](#)

DATA-CENTER ELECTRICITY

945 TWh

IEA 2030 projection, from ~ 415 TWh (2024)

IEA, Energy & AI 2025 [\[126\]](#)

EXPERT HLMI MEDIAN

2047

50% probability year, per 2,778 AI authors (2023 survey; 10% by 2027)

Grace et al. 2024 – a survey aggregate, not a fact about the future [\[127\]](#)

Two further steering currents, both measured: **geography** — China passed the US in PPP R&D (2024)^{[\[51\]](#)}, leads the Nature Index (Share 32,122 vs 22,083, 2024 data)^{[\[128\]](#)}, and entered the GII top 10 (2025)^{[\[79\]](#)}, so the next method-of-invention race is genuinely multipolar; and **energy/materials cost curves** — Wright’s-law dynamics (solar: -99.6% since 1976 at a 20.2% learning rate^{[\[95\]](#)}) remain the best-validated forecasting instrument available^{[\[93\]](#)[\[94\]](#)}, and now apply to the compute build-out itself.

Forecast humility — the base rate of confident prediction

- **FALSITY** Watson's 1943 "world market for maybe five computers" — apocryphal; no primary evidence exists (it grew from a misread 1953 remark about one machine).^[129] The most-quoted forecasting failure in tech history never happened.
- Ken Olsen's 1977 "no reason for any individual to have a computer in his home" was real but concerned home-automation mainframes, and he protested the decontextualization.^[130]
- **TRUTH** Wilbur Wright, 1908, on himself: "In 1901 I said to my brother Orville that man would not fly for fifty years. Two years later we ourselves made flights."^[131] Even the inventors misforecast their own field — in the pessimistic direction.

Hence this section's method: extrapolate measured trends, carry published ranges, name falsifiers — and treat every named-year forecast, including the survey medians above, as a claim about forecasters, not about the future.

6.5 • The cast verdict

[TRUTH] Invention's unit has moved person → lab → team for 150 years; automated search is the observed next step in the same series. [26][41][100]

[TRUTH] AI systems now produce discovery-grade outputs in narrow domains: a Nobel-recognized structure predictor, an IMO-certified gold performance, a Phase-III-stage AI-discovered drug, a record-improving algorithm search. [107][114][112][109]

[TRUTH] The human-input side plateaus this century (population peak ~2080s; researcher growth below the 13-year-doubling requirement on the per-capita series). [96][58][89]

[FALSITY] "AI has already transformed aggregate productivity." No AI-discovered drug is approved; US TFP growth had not broken its ~0.9%/yr trend through 2019, the latest series verified for this paper; the J-curve predicts exactly this lag. [113][68][85]

[IGNORANCE] Whether AI lifts research productivity enough to offset the scissors of §6.1. DATA REQUEST: BLS TFP prints 2020-2030 vs the 0.87%/yr baseline; rentosertib Phase III readout and any first FDA approval of an AI-discovered drug; METR task-length trend through 2027; replication of AI-materials utility after the Chemistry of Materials critique.

[IGNORANCE] Timing of research-capable machine intelligence.

DATA REQUEST: none can settle it now – the 2047 HLMI median is an aggregate of 2,778 forecasts, and the 13-year inward jump between the 2022 and 2023 surveys shows how unstable such medians are. Treat as sentiment data.

CONTRADICTION: Bloom et al. (research productivity falling ×41) ⊥ Census WP 25-29 (returns to R&D rising) – unresolved measurement dispute; severity moderate. Both retained. [89][91]

=> **Direction: TRUTH-grade extrapolation.** Invention continues to industrialize into automated, compute-fed search while its human-headcount input flattens; the binding constraints shift from researchers toward compute, energy, and – hardest of all – verification and selection capacity (the SELECT box of §03, which regulators, wet labs, and markets still run at human speed). **Magnitude: IGNORANCE**, bounded by the published 0.1-1.5 pp/yr range. Every premise above is falsifiable by the named data, most of it arriving before 2031.

07 · METHOD & LIMITS

How this paper knows what it claims

The apparatus, disclosed. What was verified, how, and where the edges of confidence sit.

Method

This paper was researched on **2026-09-03** by three parallel deep-research passes — (a) the history and theory of invention, (b) measurement systems and current KPI values, (c) evidence bearing on the future — each instructed to verify every claim against primary

or authoritative sources (WIPO, OECD, NSF NCSES, UN DESA, journal and publisher pages, the Nobel Foundation, Our World in Data) and to mark anything unverifiable as unusable. The assembled draft was then adversarially red-teamed claim-by-claim before release: citations spot-checked, numbers re-derived where possible, attributions tested against quote-attribution research.

Three writing rules bind every section. **Numbers carry citations** — a figure with no bracketed source does not appear. **Claims carry states** — TRUTH (verified against the cited source), FALSITY (a documented myth or overclaim), IGNORANCE (open, with an explicit request for the data that would settle it). **The future section extrapolates; it does not prophesy** — it projects only regularities that have already survived measurement, and where published expert estimates disagree, it reports the disagreement as a range rather than resolving it by preference.

Limitations — read before citing this paper onward

- **Survivorship bias is structural.** The history of invention is written from inventions that worked; the denominator of failed attempts is mostly unrecorded, so success-factor claims are weaker than they look.
- **Formal metrics undercount informal invention.** Patents and R&D budgets measure institutionalized, patentable invention; tacit know-how, process tweaks, and open-source contribution are systematically underweighted.
- **Patent counts confound output with propensity.** Differences across countries, industries, and eras partly reflect willingness and incentive to patent, not inventive activity itself.
- **Data lag.** Official global statistics run one to three years behind; “latest” figures below are the latest published, not the current instant.
- **The forecast is conditional.** Section 06 is an extrapolation of measured trends plus published estimate ranges. Confidence decays with horizon; every stated projection names its condition. Nothing in it is a prediction of specific events.
- **Source-access bias.** English-language and open-access sources are overrepresented; some primary datasets sit behind paywalls and were verified through their publishers’ own summaries.

Primary and authoritative sources, as verified on 2026-09-03. URLs are given in full so every claim can be re-checked.

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